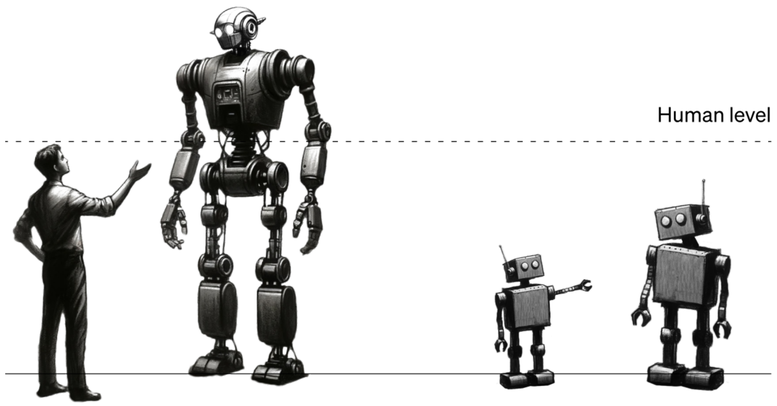
Relating Misfit to Gain in Weak-to-Strong Generalization Beyond the Squared Loss

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Weak-to-Strong Generalization



Superalignment
Supervisor: Human
Student: Superhuman AI

Analogy
Supervisor: Weak Model (GPT-2)
Student: Strong Model (GPT-4)

Methodology

1. Finetune weak model (GPT-2) on ground-truth labels (weak performance)
2. Finetune strong model (GPT-4) on *data labeled by weak model* (weak-to-strong performance)
3. Compare to strong model finetuned on ground-truth labels (strong ceiling performance)

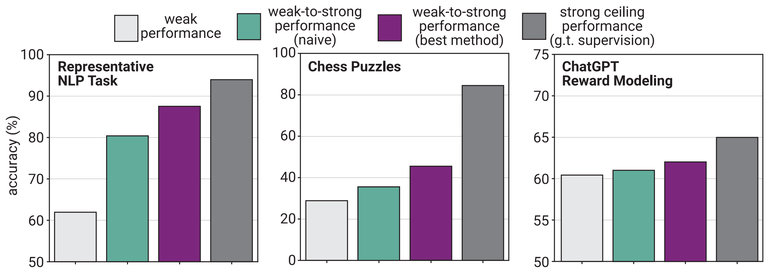


Figure 1. Weak-to-strong generalization occurs across many different problem domains[1].

Mathematical Formulation

- $\mathcal{Y}$-valued target function, for convex, open $\mathcal{Y}$:
$$f^* \circ h^*$$
- Weak model representation h_w (e.g. pretrained transformer layers of GPT-2)
- Strong model representation h_s (e.g. pretrained transformer layers of GPT-4)
- Finetuning weak model on ground-truth labels (e.g. train logistic regression layer on top of h_w) yields
$$f_w \circ h_w$$
- Finetuning strong model on weakly-supervised data (e.g. train logistic regression layer on top of h_s) yields
$$f_{sw} \circ h_s$$

Main Results for Cross-Entropy

Let $\mathcal{Y} = (0, 1)$. Let $\mathrm{xe}(p\|q)$ be the binary cross-entropy between $p, q \in \mathcal{Y}$, and $\mathrm{d}(p\|q)$ be the binary KL-divergence. Let $\mathrm{XE}(f\|g) := \mathbb{E}_x \mathrm{xe}(f(x)\|g(x))$ and $\mathrm{D}(f\|g) := \mathbb{E}_x \mathrm{d}(f(x)\|g(x))$.

Then, assuming *realizability*,

1. Suppose $\mathcal{F}_s$ is convex, and we obtain
$$f_{sw} = \arg\min_{f \in \mathcal{F}_s} \mathrm{D}(f \circ h_s \| f_w \circ h_w).$$
Then,
$$\overbrace{\mathrm{XE}(f^* \circ h^* \| f_w \circ h_w) - \mathrm{XE}(f^* \circ h^* \| f_{sw} \circ h_s)}^{\text{GAIN IN PERFORMANCE}} \geq \underbrace{\mathrm{D}(f_{sw} \circ h_s \| f_w \circ h_w)}_{\text{MISFIT}}$$
2. Otherwise, let $\mathrm{co}^k \mathcal{F}_s$ be the set of k-convex combinations of $\mathcal{F}_s$. If
$$f_{sw} = \arg\min_{f \in \mathrm{co}^k \mathcal{F}_s} \mathrm{D}(f \circ h_s \| f_w \circ h_w),$$
then (if $\mathcal{F}_s$ satisfies some regularity assumptions),
$$\mathrm{XE}(f^* \circ h^* \| f_w \circ h_w) - \mathrm{XE}(f^* \circ h^* \| f_{sw} \circ h_s) \geq \mathrm{D}(f_{sw} \circ h_s \| f_w \circ h_w) - O\left(\sqrt{1/k}\right)$$

Proof Methods

Key Tool: Pythagorean Theorem for Bregman Divergences. For C^1, strictly convex ψ on $\mathcal{Y}$, Bregman divergence of ψ is
$$d_\psi(p, q) := \psi(p) - \psi(q) - \langle \nabla\psi(q), p - q \rangle.$$
Similarly $D_\psi(f, g) := \mathbb{E}_x d_\psi(f(x), g(x))$.

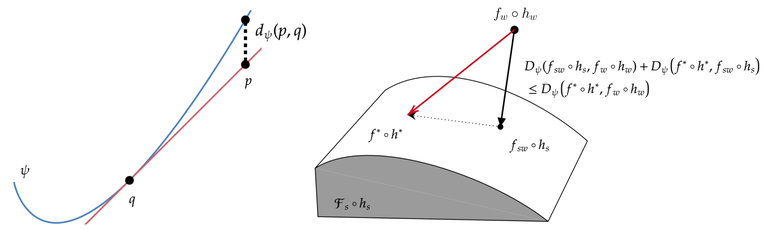


Figure 2. *Left*: Pictorial representation of a Bregman divergence. *Right*: Bregman Pythagorean Inequality, applies when $\mathcal{F}_s$ is convex.

If ψ is Shannon negentropy, $\mathrm{D} = D_\psi$, proving (1). (2) applies *probabilistic method* to show $\mathrm{co}^k \mathcal{F}_s$ approximates the convex hull $\mathrm{co}\mathcal{F}_s$.

Experimental Results

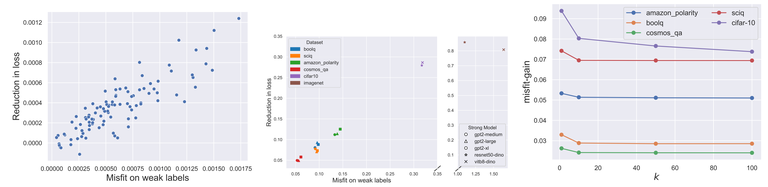


Figure 3. (1) synthetic data, (2) real-world data, (3) impact of increasing k. Inequality is evaluated w.r.t. strong ceiling.

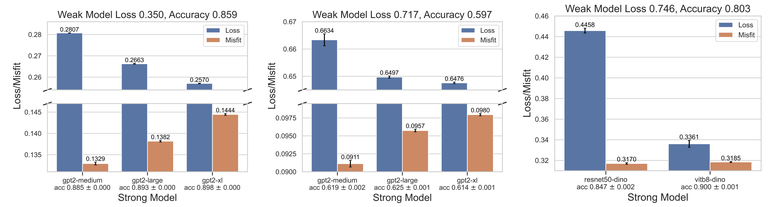


Figure 4. Impact of varying strong model. Evaluated w.r.t. ground truth. (1) Amazon Polarity (2) BoolQ (3) CIFAR-10.

[1] C. Burns, P. Izmailov, J. H. Kirchner, B. Baker, L. Gao, L. Aschenbrenner, Y. Chen, A. Ecoffet, M. Joglekar, J. Leike, I. Sutskever, and J. Wu. Weak-to-strong generalization: Eliciting strong capabilities with weak supervision, 2023.

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